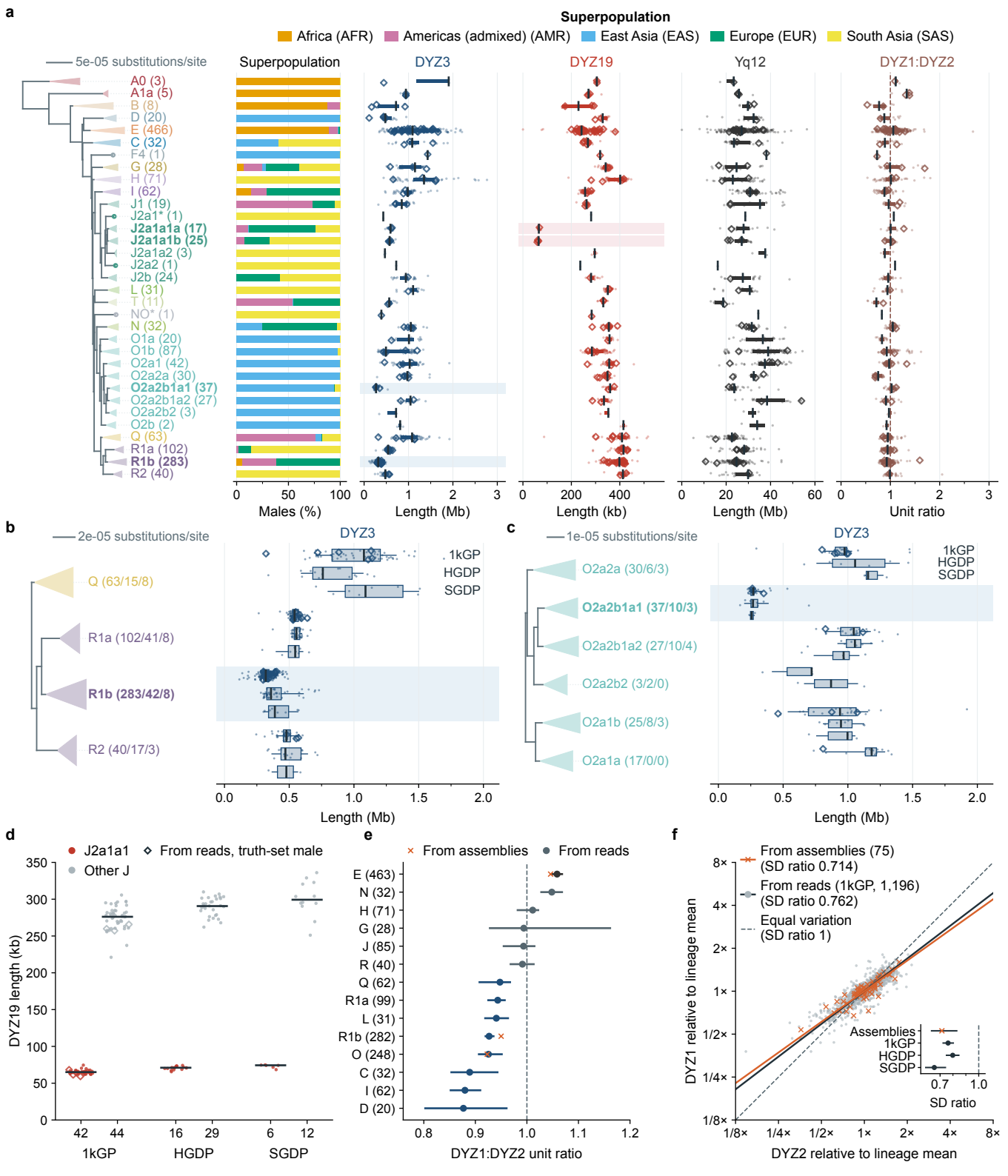

a
Superpopulation
Africa (AFR)
Americas (admixed) (AMR)
East Asia (EAS)
Europe (EUR)
South Asia (SAS)
5e-05 substitutions/site
A0 (3)
A1a (5)
B (8)
D (20)
E (466)
C (32)
F4 (1)
G (28)
H (71)
I (62)
J1 (19)
J2a1* (1)
J2a1a1a (17)
J2a1a1b (25)
J2a1a2 (3)
J2a2 (1)
J2b (24)
L (31)
T (11)
NO* (1)
N (32)
O1a (20)
O1b (87)
O2a1 (42)
O2a2a (30)
O2a2b1a1 (37)
O2a2b1a2 (27)
O2a2b2 (3)
O2b (2)
Q (63)
R1a (102)
R1b (283)
R2 (40)
Superpopulation
DYZ3
DYZ19
Yq12
DYZ1:DYZ2
0 50 100
Males (%)
0 1 2 3
Length (Mb)
0 200 400
Length (kb)
0 20 40 60
Length (Mb)
0 1 2 3
Unit ratio
b
2e-05 substitutions/site
Q (63/15/8)
R1a (102/41/8)
R1b (283/42/8)
R2 (40/17/3)
DYZ3
1kGP
HGDP
SGDP
0.0 0.5 1.0 1.5 2.0
Length (Mb)
c
1e-05 substitutions/site
O2a2a (30/6/3)
O2a2b1a1 (37/10/3)
O2a2b1a2 (27/10/4)
O2a2b2 (3/2/0)
O2a1b (25/8/3)
O2a1a (17/0/0)
DYZ3
1kGP
HGDP
SGDP
0.0 0.5 1.0 1.5 2.0
Length (Mb)
d
J2a1a1
From reads, truth-set male
Other J
DYZ19 length (kb)
350
300
250
200
150
100
50
0
42 44
16 29
6 12
1kGP
HGDP
SGDP
e
From assemblies
From reads
E (463)
N (32)
H (71)
G (28)
J (85)
R (40)
Q (62)
R1a (99)
L (31)
R1b (282)
O (248)
C (32)
I (62)
D (20)
0.8 0.9 1.0 1.1 1.2
DYZ1:DYZ2 unit ratio
f
From assemblies (75)
(SD ratio 0.714)
From reads (1kGP, 1,196)
(SD ratio 0.762)
Equal variation
(SD ratio 1)
DYZ1 relative to lineage mean
8×
4×
2×
1×
1/2×
1/4×
1/8×
Assemblies
1kGP
HGDP
SGDP
0.7 1.0
SD ratio
1/8× 1/4× 1/2× 1× 2× 4× 8×
DYZ2 relative to lineage mean